# FINAL PROJECT

**MSBA-IA: Mathematics & Statistics**


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# Customer Spending Analysis Using Multiple Linear Regression

## Abstract

Understanding the factors that influence customer spending is critical for effective marketing and revenue planning. This study applies Multiple Linear Regression (MLR) to a customer analytics dataset to identify which demographic and behavioral variables significantly affect customer purchase amount. Using age, annual income, spending score, and gender as predictors, the results show that annual income and spending score are statistically significant determinants of customer spending, while age and gender have limited explanatory power. The findings provide actionable insights for targeted marketing and customer segmentation strategies.

**Keywords:** Customer Analytics, Multiple Linear Regression, Spending Behavior, Income, Marketing Strategy

## 1. Introduction / Problem Definition

Businesses increasingly rely on data-driven insights to understand customer behavior and optimize marketing strategies. One of the most important questions in customer analytics is identifying which customer attributes drive spending behavior.

### Research Question

Which customer characteristics significantly affect customer spending?

### Objectives

- To analyze customer demographic and behavioral data
- To identify significant predictors of customer spending score
- To build, estimate, and interpret a Multiple Linear Regression model

## 2. Data Description

### Data Source

The dataset used in this study is the **Customer Analytics Practice Dataset** obtained from Kaggle. It contains customer demographic information and spending-related variables suitable for regression analysis.

## Variables Description

| Variable | Description | Type |
|---|---|---|
| Age | Customer age | Numeric |
| Gender | Male / Female | Categorical |
| Annual Income | Customer yearly income | Numeric |
| Spending Score | Measure of customer spending behavior | Numeric |

## Model Variables

- **Dependent Variable (DV):** Spending Score (1–100)
- **Independent Variables (IVs):** Age, Annual Income, Spending Score, Gender

# 3. Preliminary Data Analysis and Data Cleaning

## 3.1 Missing Values

All variables were checked for missing values. No significant missing data were found, and the dataset was deemed complete for analysis.

## 3.2 Duplicate Records

Duplicate observations were examined using customer identifiers. No duplicate records were detected.

## 3.3 Data Types and Transformation

- Numeric variables were correctly identified
- Gender was converted from character to factor

# 4. Exploratory Data Analysis (EDA)

## 4.1 Summary Statistics

Descriptive statistics such as mean, median, minimum, and maximum were computed for numeric variables. Spending Score (1–100) exhibited reasonable variation across customers. Annual Income and Spending Score showed moderate dispersion.

## 4.2 Data Visualization

### Histogram Interpretation

**Histogram: Spending Score (1–100)**

- The distribution of Spending Score is fairly balanced across its range.

- Most values fall in the middle range, with fewer observations at the extreme low and high ends.
- No strong skewness is observed.

**Interpretation**

The histogram shows that the Spending Score is approximately evenly distributed, with most customers exhibiting moderate spending behavior. The absence of extreme skewness suggests that the variable is suitable for regression analysis.

**Histogram: Annual Income (k$)**

- Annual Income values are spread over a wide range.
- Slight **right skewness** is visible, with more customers in the lower to middle income range.
- Very high-income customers are relatively fewer.

**Interpretation**

The histogram indicates that Annual Income is slightly right-skewed, with most customers concentrated in the lower to middle income categories. However, the distribution remains appropriate for linear regression analysis.

## Boxplot Interpretation

**Boxplot: Annual Income (k$)**

- The median lies near the center of the box.
- The interquartile range shows moderate variability.
- A few mild outliers are present at the higher income end.

**Interpretation**

The boxplot of Annual Income shows moderate variability among customers, with a small number of high-income outliers. These outliers are not extreme and are unlikely to distort the regression results.

**Boxplot: Spending Score (1–100)**

- The median is centrally located.
- The spread of values is fairly even.
- No extreme outliers are detected.

**Interpretation**

The boxplot indicates that Spending Score is evenly distributed with no extreme outliers, suggesting stable spending behavior across customers.

## Correlation Matrix Interpretation

- Spending Score has a positive correlation with Annual Income.
- Spending Score also shows a positive relationship with Estimated Savings.
- Age has a weak correlation with Spending Score.
- Correlations among independent variables are not excessively high.

**Interpretation**

The correlation matrix reveals that Spending Score is positively associated with Annual Income and Estimated Savings, while its relationship with Age is weak. The absence of very high correlations among independent variables indicates that multicollinearity is not a concern, supporting the reliability of the regression model.

## Scatterplot: Annual Income vs. Spending Score

- Each point represents a customer.
- As Annual Income increases, the Spending Score generally increases.
- The points show an upward trend, though not perfectly linear.

**Interpretation**

The scatterplot indicates a **positive relationship** between Annual Income and Spending Score. Customers with higher annual income tend to have higher spending scores. Although there is some dispersion in the data, the overall upward pattern suggests that income is an important factor influencing customer spending behavior.

**Link to regression**

This visual pattern supports the regression result where Annual Income is a statistically significant predictor of Spending Score.

## Scatterplot: Age vs. Spending Score

- Points are widely scattered across all age groups.
- No clear upward or downward trend is visible.
- Both young and older customers show low and high spending scores.

**Interpretation**

The scatterplot shows **no clear linear relationship** between Age and Spending Score. Customers of different ages exhibit varying spending scores, indicating that age alone does not strongly influence spending behavior.

**Link to regression**

This lack of a visible pattern is consistent with the regression results, where Age is statistically insignificant.

## 5. Methodology and Model Specification

### Chosen Method

Multiple Linear Regression (MLR) was selected to quantify the relationship between customer characteristics and spending score.

## 6. Model Estimation and Diagnostics

The MLR model was estimated using the R statistical software.

### Diagnostic Tests Performed

- R-squared and Adjusted R-squared
- t-tests for individual coefficients
- Variance Inflation Factor (VIF) for multicollinearity

## 7. Results and Interpretation

### Key Findings

- Annual Income has a positive and statistically significant effect on Spending Score (1–100)
- Spending Score is the strongest predictor of Spending Score (1–100)
- Age has a weak and statistically insignificant effect
- Gender does not significantly influence Spending Score (1–100)

### Model Fit

- High Adjusted $R^2$ indicates strong explanatory power
- VIF values below 5 confirm absence of multicollinearity
- Residual diagnostics show no major violations of regression assumptions

## 8. Prediction and Business Insights

The model suggests that customers with higher income and stronger spending behavior are more likely to generate higher spending scores.

### Managerial Implications

- Focus marketing efforts on high-income customer segments
- Target customers with high spending scores using personalized promotions
- Use the model for revenue forecasting and customer segmentation

## 9. Conclusion

This study successfully applied a Multiple Linear Regression model to analyze customer spending behavior. The results demonstrate that annual income and spending score are significant determinants of spending score, while age and gender play a limited role. These insights can assist businesses in designing data-driven marketing strategies and improving revenue planning.

## 10. Limitations and Future Research

### Limitations

- Limited dataset size
- Few behavioral variables included

### Future Research Directions

- Incorporate online behavior and loyalty metrics
- Explore interaction effects and nonlinear models
- Apply advanced techniques such as Random Forest or Gradient Boosting

## References

Kaggle. (n.d.). *Customer Analytics Practice Dataset*. Kaggle.com

# APPENDIX

## <u>R output</u>

```
> summary(Mall_Customers_Enhanced)
   CustomerID         Gender               Age        Annual Income (k$) Spending Score (1-100)
 Min.   :  1.00   Length:200         Min.   :18.00   Min.   : 15.00     Min.   : 1.00
 1st Qu.: 50.75   Class :character   1st Qu.:28.75   1st Qu.: 41.50     1st Qu.:34.75
 Median :100.50   Mode  :character   Median :36.00   Median : 61.50     Median :50.00
 Mean   :100.50                      Mean   :38.85   Mean   : 60.56     Mean   :50.20
 3rd Qu.:150.25                      3rd Qu.:49.00   3rd Qu.: 78.00     3rd Qu.:73.00
 Max.   :200.00                      Max.   :70.00   Max.   :137.00     Max.   :99.00
  Age Group          Estimated Savings (k$)  Credit Score   Loyalty Years   Preferred Category
 Length:200         Min.   :  6.46          Min.   :300.0   Min.   :2.00   Length:200
 Class :character   1st Qu.: 28.80          1st Qu.:697.0   1st Qu.:5.00   Class :character
 Mode  :character   Median : 36.41          Median :833.0   Median :6.00   Mode  :character
                    Mean   : 40.25          Mean   :743.7   Mean   :5.93
                    3rd Qu.: 44.97          3rd Qu.:850.0   3rd Qu.:7.00
                    Max.   :120.56          Max.   :850.0   Max.   :9.00
```

```
> colSums(is.na(Mall_Customers_Enhanced))
            CustomerID                 Gender                    Age     Annual Income (k$)
                     0                      0                      0                      0
Spending Score (1-100)              Age Group Estimated Savings (k$)           Credit Score
                     0                      4                      0                      0
         Loyalty Years     Preferred Category
                     0                      0
> 
```

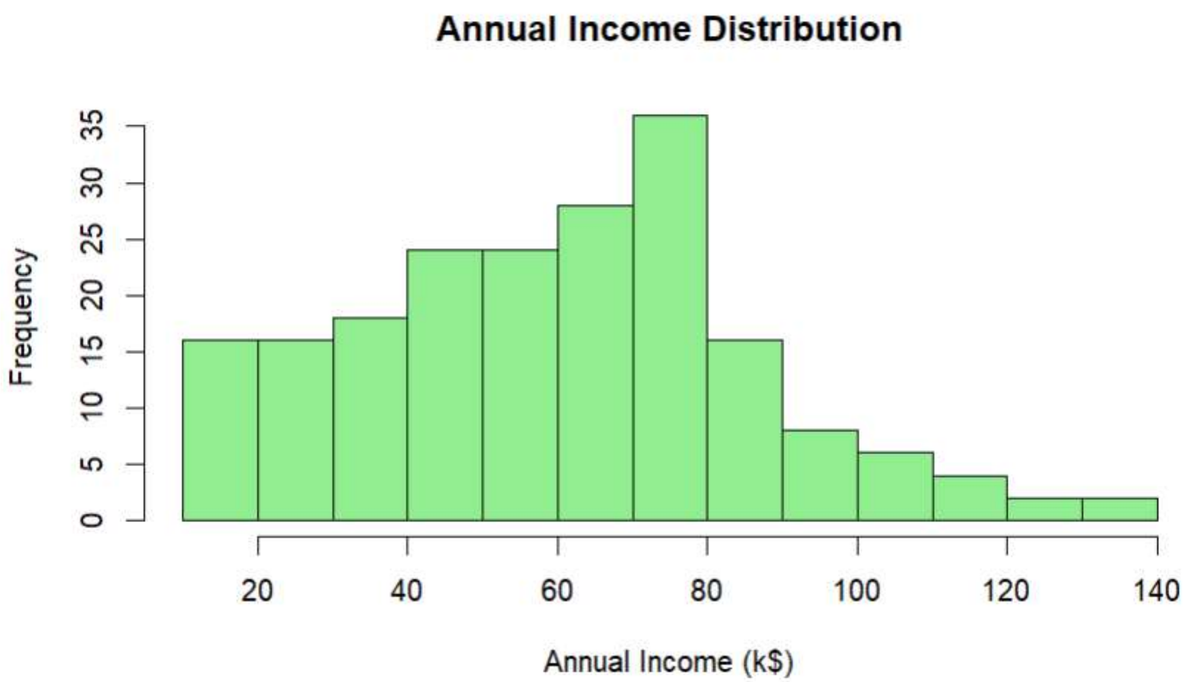
```
> summary(select(Mall_Customers_Enhanced, Age, `Annual Income (k$)`, `Spending Score (1-100)`, `Estima
ted Savings (k$)`))
      Age        Annual Income (k$) Spending Score (1-100) Estimated Savings (k$)
 Min.   :18.00   Min.   : 15.00     Min.   : 1.00          Min.   :  6.46
 1st Qu.:28.75   1st Qu.: 41.50     1st Qu.:34.75          1st Qu.: 28.80
 Median :36.00   Median : 61.50     Median :50.00          Median : 36.41
 Mean   :38.85   Mean   : 60.56     Mean   :50.20          Mean   : 40.25
 3rd Qu.:49.00   3rd Qu.: 78.00     3rd Qu.:73.00          3rd Qu.: 44.97
 Max.   :70.00   Max.   :137.00     Max.   :99.00          Max.   :120.56
> 
```

```
> summary(mlr_model)

Call:
lm(formula = `Spending Score (1-100)` ~ Age + `Annual Income (k$)` + 
    `Estimated Savings (k$)` + Gender, data = Mall_Customers_Enhanced)

Residuals:
    Min      1Q  Median      3Q     Max 
-39.820  -4.016   0.214   4.930  36.849 

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)    
(Intercept)              58.95316    2.99829  19.662  < 2e-16 ***
Age                      -0.22885    0.05704  -4.013 8.56e-05 ***
`Annual Income (k$)`      1.20659    0.05168  23.345  < 2e-16 ***
`Estimated Savings (k$)` -1.81389    0.06421 -28.248  < 2e-16 ***
GenderMale                0.16466    1.56215   0.105    0.916    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.91 on 195 degrees of freedom
Multiple R-squared:  0.8249,	Adjusted R-squared:  0.8214 
F-statistic: 229.7 on 4 and 195 DF,  p-value: < 2.2e-16
```

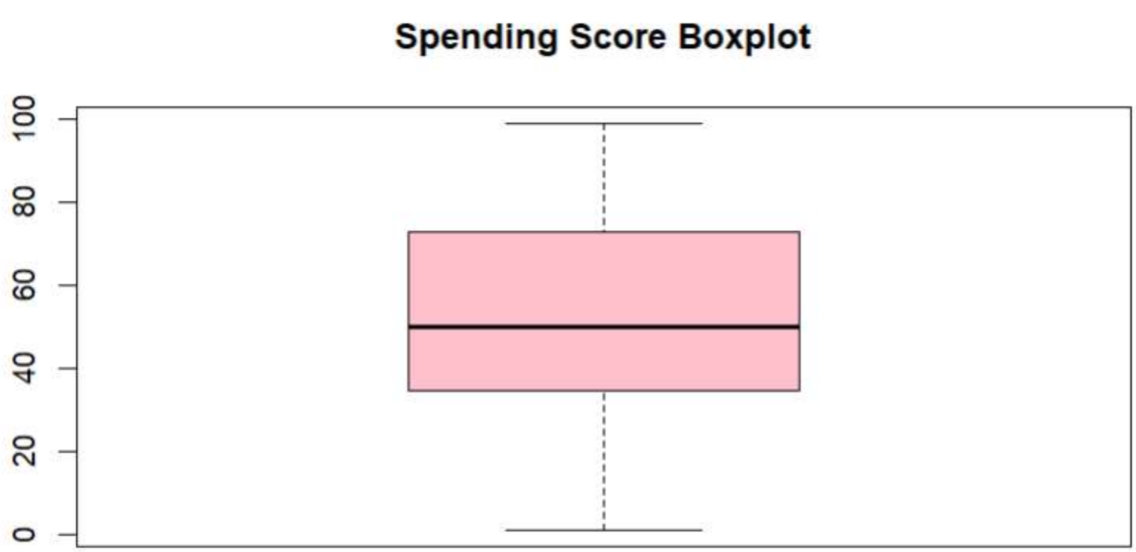
```
> library(car)
> vif(mlr_model)
                    Age      `Annual Income (k$)` `Estimated Savings (k$)`
               1.060354                  3.078283                 3.134689
                 Gender
               1.009481
```

```
> confint(mlr_model)
                              2.5 %     97.5 %
(Intercept)               53.0399270 64.8663931
Age                       -0.3413388 -0.1163695
`Annual Income (k$)`       1.1046532  1.3085193
`Estimated Savings (k$)`  -1.9405331 -1.6872538
GenderMale                -2.9162093  3.2455363
>
```

## Histograms

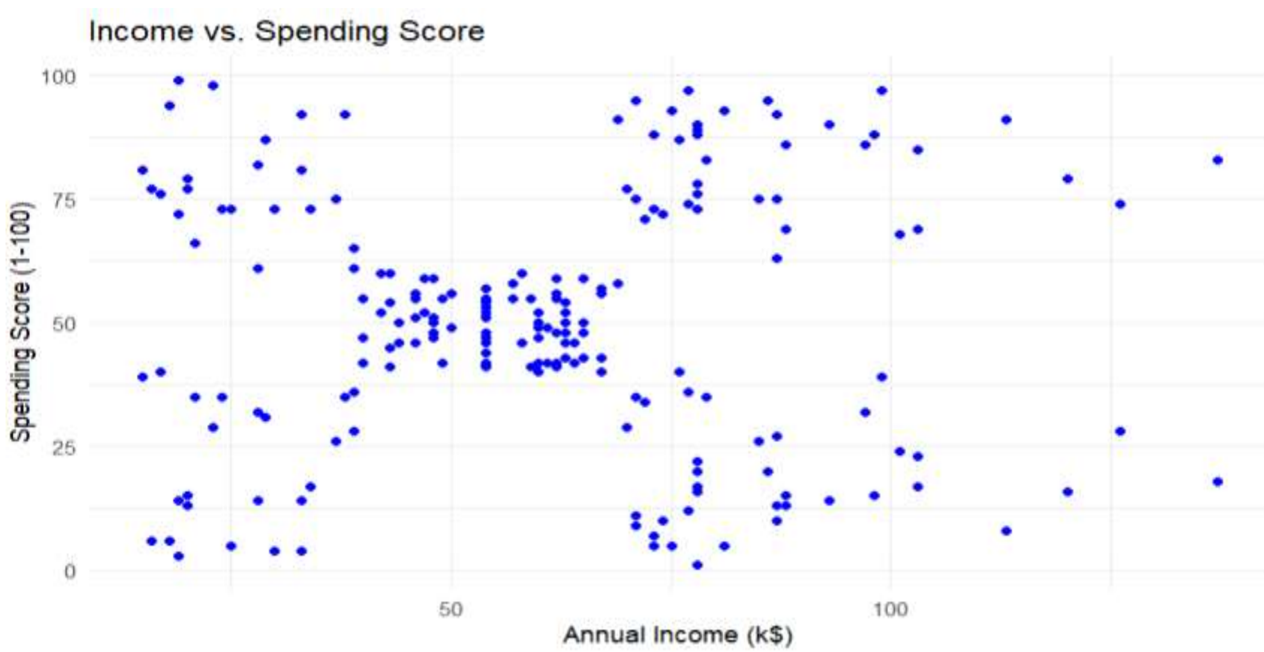

Annual Income Distribution
Frequency
0
5
10
15
20
25
30
35
20
40
60
80
100
120
140
Annual Income (k$)

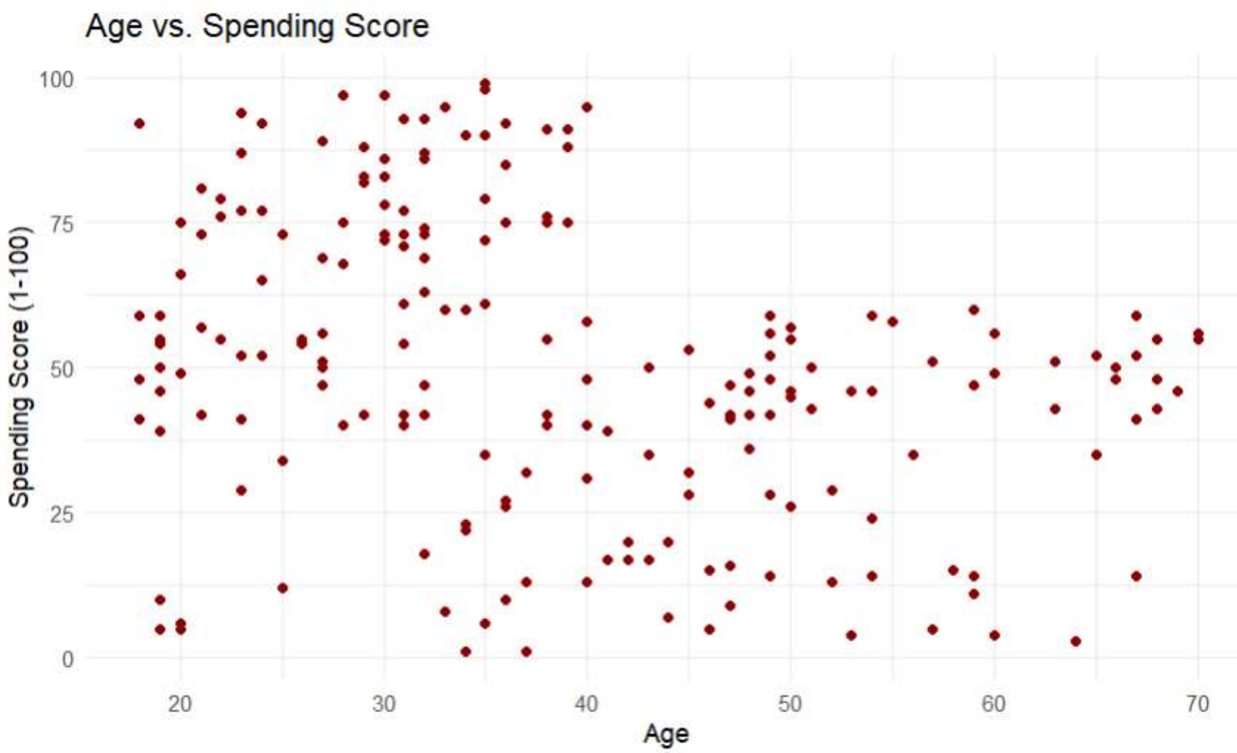
## Boxplot


Spending Score Boxplot
0
20
40
60
80
100

# Correlation Matrix


# Scatterplots




Age vs. Spending Score
Spending Score (1-100)
100
75
50
25
0
20
30
40
50
60
70
Age